

Christdev Digital Academy HTML - Foundation of Web Development

HTML - 6 hours

Course Overview:

This HTML course introduces the fundamentals of web development using HTML (HyperText Markup Language). The lessons are designed to provide students with the skills to create and structure web pages. By the end of the course, students will have a solid understanding of HTML, enabling them to build simple, functional web pages.

Course Requirements & Setup

- Download and install **Visual Studio Code (VSCode)** for HTML development
- Use any modern web browser (Chrome, Firefox, Edge) to preview HTML pages
- Install Emmet in VSCode for code expansion

Learning Outcomes:

By the end of this course, students will be able to:

- Create basic web pages using HTML
- Structure web pages with headings, paragraphs, lists, and media
- Create forms, tables, and links
- Embed audio, video, and external content (iframes)
- Work with images, favicons, and Font Awesome icons
- Use HTML entities and emojis
- Work efficiently using Emmet in VSCode

Course Outline:

Lesson 1: Introduction to HTML (2 hours)

1. Your First Webpage (20 min)

- Introduction to HTML
- Creating an HTML file
- Basic structure of an HTML document (<!DOCTYPE html>, <html>, <head>, <body>)
- Displaying text on a webpage

2. The Structure of a Webpage (20 min)

- HTML elements, tags, and nesting elements
- Understanding block-level and inline-level elements
- Practical structure: title, headings, and paragraphs

3. Creating a Full Webpage (20 min)

- Putting together multiple HTML elements into a simple webpage
- Introduction to comments (<!-- comment -->)

4. HTML Attributes (15 min)

- Using attributes in HTML elements
- Important attributes: id, class, title
- Global attributes: lang, style, dir

5. Header Tags (10 min)

- Using heading tags from <h1> to <h6>
- Proper usage and SEO benefits of headings

6. Paragraph Tags (10 min)

- Creating paragraphs with <p>
- Using line breaks with
 and horizontal rules with <hr>

7. Formatting Text (10 min)

- Using bold () and italic (<i>) tags
- Subscript and superscript: <sub>, <sup>

8. Unordered Lists (10 min)

- Creating bullet point lists with and

9. Ordered Lists (10 min)

- Creating numbered lists with `<ol>` and `<li>`
- 10. **Definition Lists (10 min)**
 - Using `<dl>`, `<dt>`, and `<dd>` to create a glossary-like list
- 11. **HTML Emojis (10 min)**
 - How to insert emojis using HTML code (e.g., `😀`, `😄`)
- 12. **Images & Favicons (15 min)**
 - Embedding images using `<img>`
 - Key attributes: `src`, `alt`, `width`, `height`
 - Adding favicons to the browser tab using the `<link>` tag
- 13. **Working with Fonts (Font Awesome) (15 min)**
 - Introduction to Font Awesome and its usage in HTML
 - Embedding Font Awesome icons in a webpage
 - Benefits of using icon fonts for visual enhancements

Lesson 2: Advanced HTML (2 hours)

1. **Forms (20 min)**
 - Creating basic forms using `<form>`, `<input>`, `<textarea>`, `<select>`, `<button>`
 - Input types: `text`, `radio`, `checkbox`, `email`, `password`
2. **Tables (20 min)**
 - Structuring data in tables using `<table>`, `<tr>`, `<th>`, `<td>`
 - Adding table headers and spanning rows/columns with `rowspan`, `colspan`
3. **Links (20 min)**
 - Creating hyperlinks using `<a>`
 - Linking to external sites, internal pages, and anchor links
4. **HTML Entities (15 min)**
 - Inserting special characters using entities (`<`, `>`, `©`, etc.)
5. **Audio & Video (15 min)**
 - Embedding media using `<audio>` and `<video>`
 - Key attributes: `controls`, `src`, `autoplay`

6. Iframes (15 min)

- Embedding external content using <iframe>
- Adjusting iframe size with width, height

7. Working with Emmet in VSCode (15 min)

- Using Emmet shorthand to speed up HTML development
- Common Emmet abbreviations and how to expand them in VSCode

HTML COURSE NOTES

Lesson 1: Introduction to HTML (2 hours)

1.1 Your First Webpage (20 min)

What is HTML?

HTML (Hypertext Markup Language) is the standard language used to create web pages. It defines the structure of a webpage and is made up of a series of elements, which are written using tags.

- **Hypertext:** Refers to text that is linked to other text or documents.
- **Markup Language:** A system for annotating a document in a way that is distinguishable from the text itself.

History: HTML was created by Tim Berners-Lee in 1991. It became the backbone of web development as it structured content on the web.

Importance of HTML:

- It's essential for creating the structure of websites.
- Works in conjunction with CSS and JavaScript to create fully functional, visually appealing websites.
- It is platform-independent and works across all browsers.

Creating an HTML File

To create an HTML file:

1. Open a text editor like Visual Studio Code (VS Code).
2. Save the file with an .html extension, e.g., index.html.

Basic Structure of an HTML Document

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Christdev Digital Academy</title>

</head>

<body>

<h1>Boost Your Coding Efficiency with Shortcuts</h1>

<p>Christdev Digital Academy.</p>

</body>

</html>

<!DOCTYPE html>: Tells the browser that the document is written in HTML5.

<html lang="en">: The root element of the webpage, with the language attribute set to English.

<head>: Contains metadata about the webpage (e.g., character set, viewport, title).

`<meta charset="UTF-8">`: Specifies the character encoding used, ensuring the page displays special characters correctly.

`<meta name="viewport" content="width=device-width, initial-scale=1.0">`: Optimizes the page for mobile devices, making it responsive.

`<title>`: The title that appears on the browser tab.

`<body>`: Contains all the visible content, including headings, paragraphs, images, etc.

Displaying Text on a Webpage

To display text:

`<h1>Boost Your Coding Efficiency with Shortcuts</h1>`

`<p>Christdev Digital Academy is your gateway to mastering the tech world.</p>`

`<h1>` to `<h6>`: Defines headings, with `<h1>` being the most important.

`<p>`: Defines paragraphs.

1.2 The Structure of a Webpage (20 min)

HTML Elements, Tags, and Nesting

- **HTML Element**: The content between the opening and closing tags.
- **Tags**: Delimiters in the document, usually consisting of a keyword enclosed in angle brackets (`<>`).
 - Example: `<h1>Heading</h1>` (Opening tag and closing tag)
- **Nesting Elements**: Placing one element inside another. Ensure proper indentation for readability.

Block-Level vs Inline-Level Elements

- **Block-Level Elements**: Always start on a new line and take up the full width (e.g., `<div>`, `<p>`, `<h1>` - `<h6>`).

- **Inline-Level Elements:** Do not start on a new line and only take up as much space as necessary (e.g., `<span>`, `<a>`, `<em>`).

Exercise:

Create a simple webpage that displays ChristDev Digital Academy's core message using `<h1>`, `<p>`, and `<em>` to emphasize important text.

Practical Structure: Title, Headings, and Paragraphs

- **Creating a Full Webpage:**
 - Combine multiple elements to create a simple webpage.

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
    <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width,  
initial-scale=1.0">
```

```
    <title>Christdev Digital Academy</title>
```

```
</head>
```

```
<body>
```

```
    <h1>Welcome to Christdev Digital Academy</h1>
```

```
    <p>Your journey to mastering coding starts here.</p>
```

```
    <h2>Our Courses</h2>
```

```
    <p>Explore our diverse range of courses designed to boost your  
coding skills.</p>
```

```
</body>
```

```
</html>
```

Introduction to Comments:

- **Comments** are notes in the code that are not displayed on the webpage.
- Example: `<!-- This is a comment -->`

Shortcut for creating comments in VSCode: Type `<!-- comment -->` or select text and press Ctrl + / (Windows) or Cmd + / (Mac).

1.3 HTML Attributes (15 min)

Using Attributes in HTML Elements

- **Attributes** provide additional information about an element.
- Examples of important attributes:
 - id: Unique identifier for an element (e.g., `<p id="intro">`).
 - class: Specifies a class for an element (e.g., `<p class="highlight">`).
 - title: Displays additional information when hovered over (e.g., `<a title="Go to homepage">`).

Global Attributes:

- Attributes that can be applied to any HTML element:
 - lang: Specifies the language of the element (e.g., `<html lang="en">`).
 - style: Applies inline CSS styles (e.g., `<p style="color:red;">`).
 - dir: Specifies the text direction (e.g., `<p dir="rtl">`).

Exercise:

1. Create a paragraph with an id of "introduction" and a "highlight" class in your HTML file.
2. Add a title attribute to a link that leads to Christdev Digital Academy's homepage.

Exercise:

Create a webpage that displays a heading and a paragraph with class and id attributes. Hovering over the text should display a title.

1.4 Header Tags (10 min)

Using Heading Tags from <h1> to <h6>

- Headings are used to define the hierarchy of content on a webpage.
- Example:

<h1>Main Title</h1>

<h2>Sub Title</h2>

<h3>Section Title</h3>

<h4>Subsection Title</h4>

<h5>Minor Section Title</h5>

<h6>Smallest Title</h6>

- **Proper Usage and SEO Benefits of Headings:**
 - Search engines use headings to understand the content structure.
 - Use headings in a hierarchical manner (e.g., <h1> for the main title, <h2> for sections).

Exercise:

1. Create a webpage with at least three headings (<h1>, <h2>, and <h3>).
2. Explain how the structure helps with SEO.

Exercise:

Create a webpage with headings (<h1> to <h6>) that outline the syllabus for ChristDev Digital Academy.

1.5 Paragraph Tags (10 min)**Creating Paragraphs with <p>**

- Use the <p> tag to create paragraphs of text.

<p>This is a paragraph explaining the courses at Christdev Digital Academy.</p>

- **single Line Breaks:**
 -
: Inserts a line break.
 - <hr>: Creates a horizontal line.

Example:

<p>This is a paragraph.
This text is on a new line.</p>

<hr>

Exercise:

1. Write a paragraph about the importance of coding education at Christdev Digital Academy.
2. Include a line break and a horizontal rule in your paragraph.

1.6 Formatting Text (10 min)**Using Formatting Tags**

- **Bold:** `<b>` or `<strong>` makes text bold.
- **Italic:** `<i>` or `<em>` makes text italic.
- **Deleted Text:** `<del>` indicates deleted text.
- **Highlighted Text:** `<mark>` highlights text.

Example:

`<p>This text is bold, italic, and deleted.``</p>`

Subscript and Superscript:

- **Subscript:** `<sub>` places text below the baseline (e.g., H₂O).
- **Superscript:** `<sup>` places text above the baseline (e.g., E=mc²).

Exercise:

1. Format a sentence about Christdev Digital Academy using bold, italic, and highlight tags.
2. Create a subscript for water (H₂O) and a superscript for Einstein's formula (E=mc²).

Exercise:

Create a webpage with bold, italic, subscript, and superscript text explaining ChristDev Digital Academy's course structure.

Unordered Lists (10 min)

Creating Bullet Point Lists with `<ul>`

- Use `<ul>` for unordered lists and `<li>` for list items.

You can customize the bullet type with the `type` attribute (`disc`, `square`, `circle`).

Example:

Web Development: HTML, CSS, JavaScript

Mobile Development: React Native, Flutter

<u>Data Science</u>: Python, R

- **Nested Lists:** Lists within lists can be created by placing another inside an .

Example:

Programming Languages

Python

JavaScript

Exercise:

1. Create an unordered list of three skills students can learn at Christdev Digital Academy.
2. Include a nested list of programming languages under a skill.

Ordered Lists (10 min)

Creating Numbered Lists with

- Use for ordered lists where the order of items matters.
- Can be reversed using **reversed**.

Example:

```
<ol>  
  
  <li>Enroll in a course</li>  
  
  <li>Complete the assignments</li>  
  
  <li>Participate in discussions</li>  
  
</ol>
```

Changing List Types:

- You can change the list style with the type attribute:
 - type="A" for uppercase letters
 - type="a" for lowercase letters
 - type="I" for uppercase Roman numerals
 - type="i" for lowercase Roman numerals

Example:

```
<ol type="A">  
  
  <li>First Item</li>  
  
  <li>Second Item</li>  
  
</ol>
```

Exercise:

1. Create an ordered list of steps to register for a course at Christdev Digital Academy.

2. Change the list type to uppercase letters.
3. Create an ordered list of ChristDev Digital Academy's course modules in reverse order.

Links and Anchors (10 min)

Creating Links with `<a>` Tag

- Use the `<a>` tag to create hyperlinks.

Example:

```
<a href="https://www.christdev.com">Visit Christdev Digital Academy</a>
```

- **Targeting Links:** Use `target="_blank"` to open links in a new tab.

Example:

```
<a href="https://www.christdev.com" target="_blank">Open in new tab</a>
```

Creating Internal Links:

- Use anchors to link to specific sections within the same page.
- Create an anchor with an id and link to it.

Example:

```
<h2 id="courses">Our Courses</h2>
```

```
<a href="#courses">Go to Courses</a>
```

The `rel="noreferrer"` attribute in the `<a>` tag is used to enhance security and privacy when linking to external websites. It prevents the browser from sending the "referrer" information to the destination site. Here's a detailed explanation:

Understanding the `referrer`

When you click a link on a webpage, your browser typically sends **referrer information** to the destination site. This referrer tells the destination website where the traffic is coming from — specifically, the URL of the webpage you were on before clicking the link.

For example, if you're on <https://christdevdigitalacademy.com> and you click a link that takes you to another site, like <https://example.com>, the example.com server would typically receive information that you came from <https://christdevdigitalacademy.com>. This can help the target website understand its traffic sources.

What Does **noreferrer** Do?

When you use the `rel="noreferrer"` attribute in an anchor (`<a>`) tag, the browser will **not** send the referrer information to the destination website. This adds a layer of privacy, ensuring the external site doesn't know where the traffic is coming from.

```
<a href="https://example.com" rel="noreferrer">Visit Example</a>
```

In this case, if someone clicks on "Visit Example," the website <https://example.com> will **not** receive any information about where the visitor came from.

Why Use `rel="noreferrer"`?

1. **Privacy:** It prevents the destination website from seeing the referrer, which can be useful if you don't want external sites to track the users coming from your website.
2. **Security:** It can also enhance security by preventing certain types of malicious behavior, like referrer-based attacks.
3. **Linking to Untrusted Sites:** When linking to external websites that might not be fully trusted, using `noreferrer` can help mitigate the risk of any malicious activity that relies on referrer data.

Example Use Case for Christdev Digital Academy

Let's say Christdev Digital Academy links to a third-party site for additional resources, but you don't want that third-party site to know the traffic came from Christdev's website. You can use `rel="noreferrer"` to prevent sharing that information.

```
<a href="https://externalresource.com" rel="noreferrer">External  
Resource</a>
```

Now, when students click this link, `https://externalresource.com` will not know they came from `https://christdevdigitalacademy.com`.

Combining `noreferrer` with `noopener`

In many cases, `noreferrer` is often combined with `noopener`

Exercise:

1. Create a link to the homepage of Christdev Digital Academy.
2. Add an internal link to a section of your webpage.

Images and Multimedia (20 min)

Embedding Images with `<img>`

- Use the `<img>` tag to add images to your webpage.

Example:

```

```

- **Attributes:**

- src: URL of the image.
- alt: Text displayed if the image cannot load.
- width and height: Control the image size.

Embedding Videos with <video>

- Use the <video> tag to include videos.

Example:

```
<video width="320" height="240" controls>
```

```
<source src="video.mp4" type="video/mp4">
```

Your browser does not support the video tag.

```
</video>
```

Exercise:

1. Embed an image of Christdev Digital Academy's logo in your webpage.
2. Add a video tutorial link using the <video> tag.

Tables (20 min)

Creating Tables with <table>

- Use the <table> tag to create a table layout.

Example:

```
<table>
```

```
<tr>

<th>Course</th>

<th>Duration</th>

<th>Price</th>

</tr>

<tr>

<td>Web Development</td>

<td>3 months</td>

<td>$300</td>

</tr>

<tr>

<td>Data Science</td>

<td>4 months</td>

<td>$400</td>

</tr>

</table>
```

- **Table Structure:**

- `<table>`: Defines the table.
- `<tr>`: Table row.
- `<th>`: Table header cell (bold and centered by default).
- `<td>`: Table data cell.

Styling Tables:

- Use CSS for better table presentation (covered in future lessons).

Spanning Rows with rowspan

- **Rowspan:** The rowspan attribute allows a cell to span multiple rows.

Example of Rowspan

```
<table border="1">
```

```
  <tr>
```

```
    <th>Course Name</th>
```

```
    <th>Duration</th>
```

```
    <th>Price</th>
```

```
  </tr>
```

```
  <tr>
```

```
    <td rowspan="2">Web Development</td> <!-- This cell spans two
rows -->
```

```
    <td>3 months</td>
```

```
    <td>$300</td>
```

```
  </tr>
```

```
  <tr>
```

```
    <td>Full Stack Development</td>
```

```
    <td>$500</td>
```

```
  </tr>
```

```
  <tr>
```

```
    <td>Data Science</td>
```

```
    <td>4 months</td>
```

```
<td>$400</td>
```

```
</tr>
```

```
</table>
```

Spanning Columns with colspan

- **Colspan:** The colspan attribute allows a cell to span multiple columns.

Example of Colspan

```
<table border="1">
```

```
<tr>
```

```
<th colspan="3">Courses Offered at Christdev Digital Academy</th>
```

```
<!-- This header spans three columns -->
```

```
</tr>
```

```
<tr>
```

```
<th>Course Name</th>
```

```
<th>Duration</th>
```

```
<th>Price</th>
```

```
</tr>
```

```
<tr>
```

```
<td>Web Development</td>
```

```
<td>3 months</td>
```

```
<td>$300</td>
```

```
</tr>
```

```

<tr>
  <td>Data Science</td>
  <td>4 months</td>
  <td>$400</td>
</tr>

<tr>
  <td colspan="2">Total Courses Offered</td> <!-- This cell spans two
columns -->
  <td>2</td>
</tr>

</table>

```

Combining Rowspan and Colspan

You can combine both rowspan and colspan in a single table to create complex layouts.

Example Combining Rowspan and Colspan

```

<table border="1">
  <tr>
    <th colspan="3">Courses Offered at Christdev Digital Academy</th>
  </tr>

  <tr>
    <th>Course Name</th>
    <th>Duration</th>
    <th></th>
  </tr>

```

<th>Price</th>

</tr>

<tr>

<td rowspan="2">Web Development</td>

<td>3 months</td>

<td>\$300</td>

</tr>

<tr>

<td>Full Stack Development</td>

<td>\$500</td>

</tr>

<tr>

<td>Data Science</td>

<td>4 months</td>

<td>\$400</td>

</tr>

<tr>

<td colspan="2">Total Courses Offered</td>

<td>3</td>

</tr>

</table>

Exercise:

1. Create a table for the courses offered at Christdev Digital Academy, including course name, duration, and price.
2. Add a header row and at least three courses.

```
<table border="1">
  <tr>
    <th>Course Name</th>
    <th>Duration</th>
    <th>Price</th>
  </tr>
  <tr>
    <td>Web Development</td>
    <td>3 months</td>
    <td>$300</td>
  </tr>
  <tr>
```

```
<td>Data Science</td>

<td>4 months</td>

<td>$400</td>

</tr>

</table>
```

Semantic HTML (20 min)

Understanding Semantic HTML

- **What is Semantic HTML?** Elements that clearly describe their meaning to both the browser and the developer.
- **Examples of Semantic Elements:**
 - `<header>`: Represents the introductory content of a page or section.
 - `<nav>`: Contains navigation links.
 - `<main>`: The main content area of the document.
 - `<article>`: Represents a self-contained composition.
 - `<footer>`: Contains footer information.

Example:

```
<header>

<h1>Welcome to Christdev Digital Academy</h1>

<nav>

<ul>

<li><a href="#courses">Courses</a></li>

<li><a href="#contact">Contact Us</a></li>

</ul>
```

```
        </nav>
</header>
<main>
    <article>
        <h2>Our Courses</h2>
        <p>Learn various programming languages...</p>
    </article>
</main>
<footer>
    <p>&copy; 2024 Christdev Digital Academy</p>
</footer>
```

Exercise:

1. Structure your webpage using semantic HTML elements like `<header>`, `<nav>`, `<main>`, and `<footer>`.
2. Create a navigation menu in the header.

Forms (30 min)

Creating Forms with `<form>`

- Use the `<form>` tag to collect user input.

Example:

html

Copy code

```
<form action="submit_form.php" method="POST">

    <label for="name">Name:</label>

    <input type="text" id="name" name="name" required><br><br>

    <label for="email">Email:</label>

    <input type="email" id="email" name="email" required><br><br>

    <input type="submit" value="Submit">

</form>
```

- **Form Attributes:**

- action: URL to send form data.
- method: HTTP method (GET or POST).

Form Elements:

- <input>: Different types like text, email, password, radio, checkbox.
- <textarea>: For multi-line text input.
- <select>: Dropdown list.

Example of a Dropdown:

```
<label for="course">Choose a course:</label>

<select id="course" name="course">

    <option value="web">Web Development</option>

    <option value="data">Data Science</option>

</select>
```

Exercise:

1. Create a contact form for Christdev Digital Academy with fields for name, email, and course selection.

2. Include a submit button.

Take-Home Assignments and Exercises

Lesson 1 Assignments:

1. Exercise 1: Your First Webpage

- **Task:** Create a simple webpage for **ChristDev Digital Academy** with:
 - A title, a heading, and a paragraph introducing ChristDev.
 - A section that includes an unordered list of the services offered by ChristDev.
 - A definition list explaining key terms like "Web Development," "Software Development," and "IT Consultancy."
- **Deliverables:** Submit the HTML file with proper structure and text formatting.

2. Exercise 2: Images and Favicons

- **Task:** Add an image of ChristDev's logo to the webpage created in Exercise 1. Include a favicon for the webpage using the <link> tag in the <head> section.
- **Deliverables:** Submit the updated HTML file with the image and favicon.

Lesson 2 Assignments:

1. Exercise 1: Forms and Tables

- **Task:** Create a contact form for **ChristDev Digital Academy** that includes:
 - Fields for name, email, phone number, and a message box.
 - A submit button that does not yet connect to a backend.

- Additionally, create a table that displays available courses, fees, and duration.
- **Deliverables:** Submit the HTML file containing the form and table.

2. Exercise 2: Links and Iframes

- **Task:** Create a webpage with:
 - Links to external resources (e.g., ChristDev's social media pages).
 - An embedded iframe that shows a YouTube video about ChristDev Digital Academy.
- **Deliverables:** Submit the HTML file with links and iframes.

3. Exercise 3: Using Emmet in VSCode

- **Task:** Recreate the basic structure of a webpage using Emmet abbreviations in VSCode. Use Emmet to quickly create a form, table, and list.
- **Deliverables:** Submit the HTML file created using Emmet.

Projects:

1. Project 1: Build a Simple Website for ChristDev Digital Academy

- **Task:** Build a website with three pages:
 - **Home page:** Includes a heading, introductory text, an image, and an ordered list of courses.
 - **Courses page:** Contains a table listing the available courses, fees, and durations.
 - **Contact page:** Includes a form where potential students can fill out their details to inquire about the courses.
- **Deliverables:** Submit the HTML files for all three pages.

2. Project 2: Student Registration Form

- **Task:** Create a registration form for ChristDev Digital Academy's students. Include fields for full name, date of birth, email, selected course, and additional comments.
- **Deliverables:** Submit the HTML file with the fully functional form structure.